Minos

Physically grounded scientific environments at compute speed

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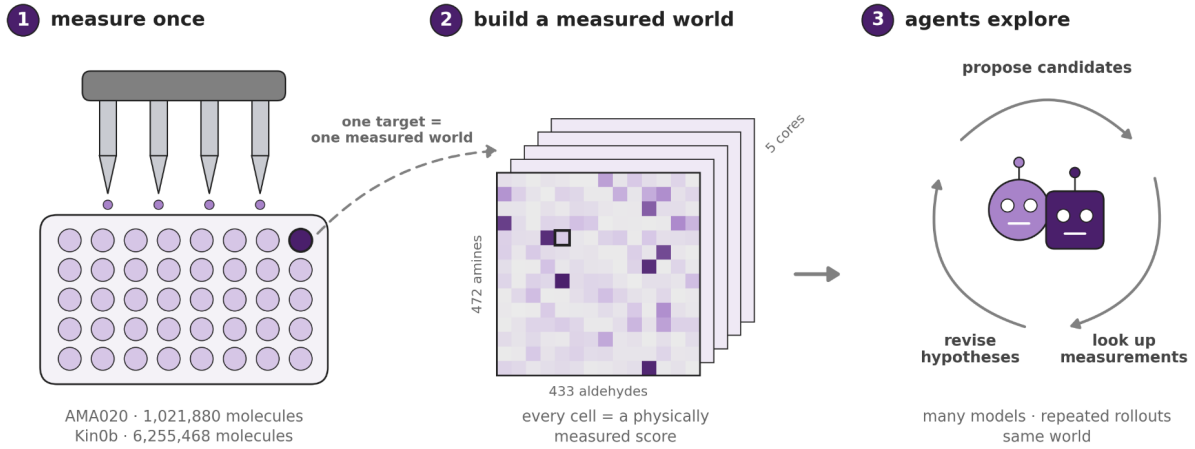
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Resources: strelka.leash.bio

ABSTRACT

Experience is critical to learning. The learner must be able to make sequential choices and receive their consequences. Scaled experience in synthetic or simulated environments has driven progress in computational learning, yet experience obtained in the physical world does not scale easily. One solution is to use laboratory techniques to premeasure a large action space and instruct agents to iteratively explore it, allowing them to make choices, receive feedback, and update their hypotheses and decisions. To embody this idea, we physically measured large spaces of small molecule-protein binding interactions converted into replayable environments. Inside these environments, named Minos worlds, scientific agents essentially run lab-in-the-loop campaigns at compute speed: they select experiments within constrained budgets, observe measured physical outcomes, and adapt their subsequent choices. Across two independently constructed chemical libraries, these environments stratified models and effort configurations by performance; inference effort increases frequently led to better outcomes. Agents depended on molecular structure to make meaningful progress. Possessing the results of all choices agents did not make but could have — counterfactual density — allowed us to compare how different agents searched, stopped, and decided when to pursue local chemical neighborhoods. We release STRELKA, a subset of these environments and associated agent trajectories for further study of scientific agents, available at strelka.leash.bio.



INTRODUCTION

Scientific discovery is a sequential decision-making problem. Scientists cannot think their way into accurate descriptions of the world. Instead, they form hypotheses about an initially uncertain physical system, decide how to interact with that system, observe the result, and use that evidence to choose the next experiment. An effective scientist — or scientific agent — must therefore do more than know facts or make accurate predictions: they must use a limited experimental budget to learn about a hidden physical world. Exploration of such worlds with limited budgets is at the heart of all technological endeavors, from materials science to medicine.

In order to build intuition on what experimental choices to make, a scientist learns what approaches are likely to work and likely to fail through experience [1, 2]. Agentic scientists can gain that experience at a superhuman scale. They can do this by interrogating legacy datasets from the literature, a rapid process but one biased by the human choices that went into generating those legacy datasets [3], or they may gain experience with simulations, which are imperfect representations of physical reality [4]. Alternatively agents may gain experience by directing physical lab-in-the-loop experiments, which allow for more potential choices but can be constrained by the lengthy cycle times — potentially weeks or months — each iteration of the loop demands [5].

Each of these approaches to gain scientific experience has limitations. To address them, we sought to reproduce the open-ended decision structure of a lab-in-the-loop without the latency between experimental cycles. We realized that densely measured experimental datasets could provide this structure: by measuring a sufficiently large fraction of the action space in advance, experimental cost can be paid up front and amortized across future agent runs. We call this framework of premeasured experimental worlds Minos.

Minos is a collection of replayable worlds in which agents can search for experimental choices, act, receive outcomes, update hypotheses, and act again. The vast majority of the action space is hidden from agents; they must pursue objectives on limited experimental budgets. Crucially, Minos is physically grounded: outcomes returned to agents derive from wet-lab measurements rather than human or model judgments. Here, we describe Minos worlds engineered to allow agents to perform small molecule structure-activity relationship (SAR) campaigns.

SAR in medicinal chemistry is critical for advancing small molecule therapies to patients. In this process, small molecules interacting with protein targets are refined over a number of experimental cycles: various chemical fragments are substituted in order to improve activity, selectivity, and improve druglike properties [6]. These cycles can take months or even years. The SAR tasks described here are solely focused on improving binding of small molecules to their respective protein targets.

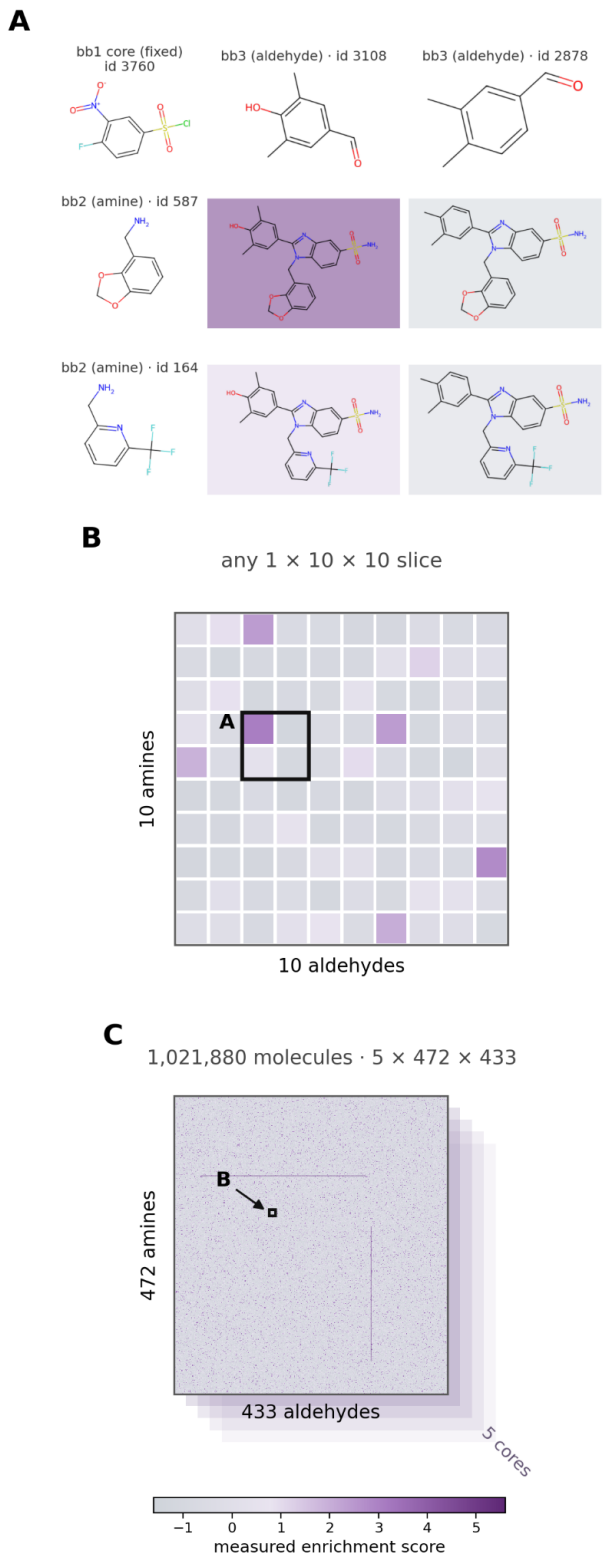


Figure 1. Minos environments are replayable experimental worlds. A) Four molecules from AMA020: all share a common core (top left) and are combinatorially constructed with four other building blocks. Binding enrichment for this target is proportional to shading. B) Expanded cartoon showing 100 molecules and illustrated binding enrichment. C) Expanded cartoon approximating the true size of a single AMA020 environment. True AMA020 called hit rate is approximately 0.238% of molecules tested.

INTRODUCTION / CONTINUED

Given a scientific objective like SAR optimization, agents are introduced to a Minos environment and allowed to perform their own experiments to interrogate a hidden world. This environment contains not only the decision trajectories made by these agents, but dense counterfactual ground truth around those trajectories. With Minos, we are able to evaluate not only every decision an agent has made, but compare them to every decision the agent could have made but did not.

We show such worlds are able to separate models and model effort levels by performance on an SAR task and token budgets. Beyond the outcomes of the tasks — whether agents were able to identify promising chemical material for a given target — we interrogate model decision processes, finding a suite of strategies employed by stronger models that are underused by weaker ones.

Accompanying the general description of Minos, we release the Scientific Testbed for Reasoning, Experimentation, Learning, Knowledge, and Agency (STRELKA), a collection of Minos worlds composed of some 34 protein targets screened against AMA020 as Harbor packages [7]. We also release a collection of some 4,602 SAR trajectory narrations produced by a variety of large language models traversing STRELKA. Please visit strelka.leash.bio for more information.

METHODS

Minos world construction

Minos environments in this work are composed of a single protein target and known results of all binding tests of molecules in a dense chemical library (Figure 1). The data corpus populating different Minos collections is composed of one of two DNA-encoded chemical libraries (DELs, Figure 3). AMA020 was assembled using a design based on Wellaway et al. [8], by sequential diversification of DNA-linked ortho-halo nitroarenes with primary amines, followed by nitro reduction and benzimidazole-forming cyclization with aldehydes. Kin0b was constructed as a 3-cycle library with 36 cores connected to building blocks with Suzuki or amide coupling; kin0b has exceptional counterfactual density.

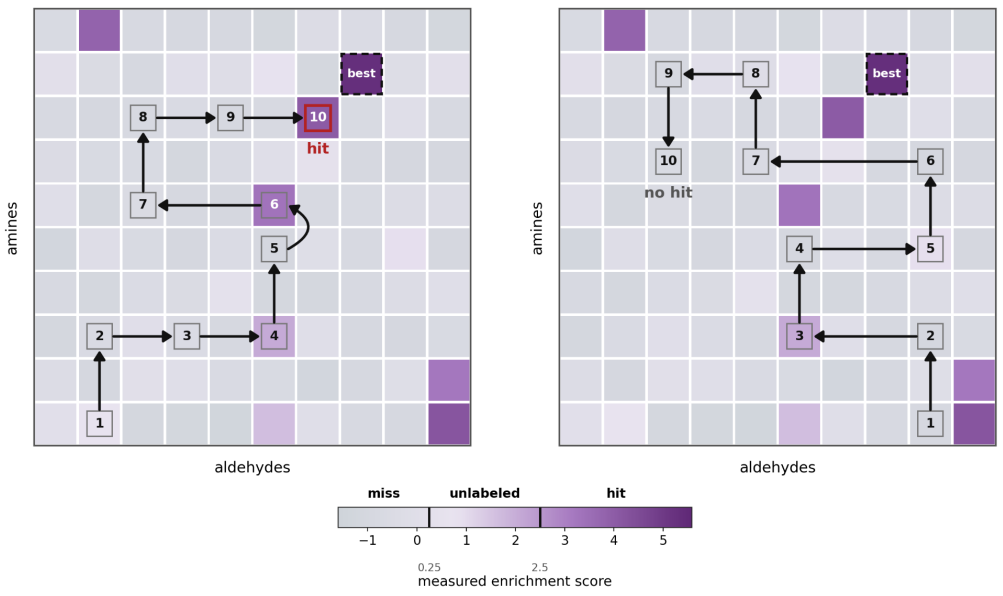


Figure 2. Left panel, a cartoon of a decision path through a Minos environment resulting in a hit. Right panel, a cartoon of a decision path through a Minos environment resulting in no hits. Best hit in the environment label went undiscovered.

Protein targets manufactured in-house were screened against these chemical libraries as described [9] in our laboratory. Total screens to date are roughly 350 proteins against AMA020 and 1,100 proteins against kin0b; analyses here are confined to 34 targets against AMA020 and 25 against kin0b (Figure 3).

We scored molecules with a proprietary enrichment algorithm and assigned each one of three binding classes (“bind”, “miss”, and “unlabeled”, the latter being too close to call, Figure 2). We selected these based on hit frequency (>200 hits, <1% of total molecules), target diversity, and task difficulty.

METHODS / CONTINUED

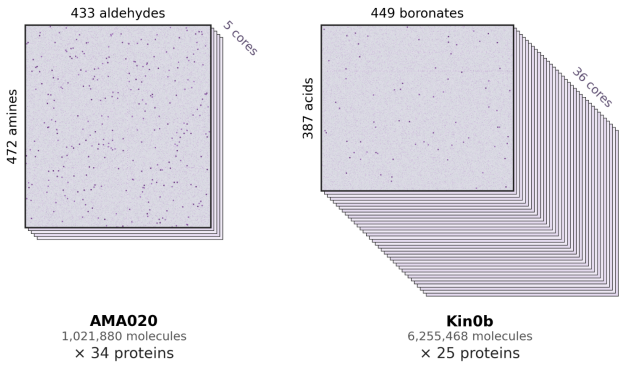


Figure 3. **Scale of Minos environments examined in this work.**

The basic Minos task

In a task designed to resemble an SAR campaign to discover more potent chemical material, agents are instructed to find the molecule with the best enrichment score against a single target out of a search space of ~1M molecules (in AMA020) or ~6M molecules (in kin0b) on a limited experimental budget. They search the orderable ~1M or ~6M collection for chemical motifs of interest, predict binding to the target of molecules containing those motifs, and experimentally test those predictions by interrogating premeasured physical outcomes (Figure 4). While SAR campaigns in classical drug discovery include optimization across target potency, selectivity, and a number of ADMET properties, we use the term here to denote increases in binding enrichment on a single target only.

We focus on one task deployed across many targets: given a target, find the best binder out of the entire chemical library (~1M or ~6M) in an experimental budget of 100 molecules. From premeasured laboratory experiments, we assigned enrichment scores to each protein-ligand interaction (PLI) using a proprietary scoring method that allows us to identify molecules that are enriched by the target relative to various control conditions. Agents spend their budgets on up to 5 molecules at a time and use the results of experimental tests on those molecules to inform future choices. Each agent-inference effort configuration conducted 10 trials within a single protein target × chemical library environment. This is a cold-start task, requiring agents to identify binders to a given protein target with no starting points: they are provided the target and the catalog of small molecules only.

To perform the task, agents are provided 3 MCP tools: `search_building_blocks`, which is a SMARTS [10] query of building blocks that make up library members; `synthesize`, which the agent uses to combine the building blocks into a molecule and queries the premeasured data for its enrichment; and `get_status`, which keeps a tally of progress. The full prompt is available in the Appendix (Cold start prompt). All transcripts are available at strelka.leash.bio.

To understand the contributions of model family and effort level, we tested 58 different chemical library × model × effort level combinations.

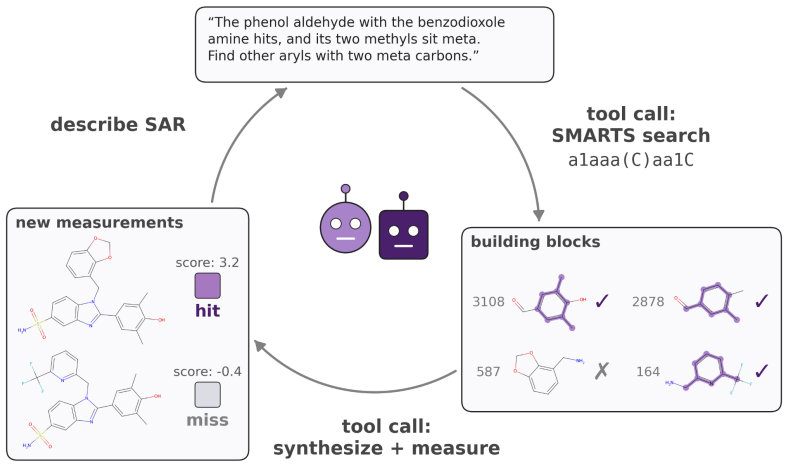


Figure 4. **The Minos experimental cycle.** Agents propose SAR hypotheses, search for building blocks in our chemical libraries to test those hypotheses using SMARTS [10], computationally assemble molecules from those building blocks using reaction SMARTS [11] and receive binding scores for them by querying premeasured data.

RESULTS

A Minos collection of experimental worlds stratifies models by capability and effort

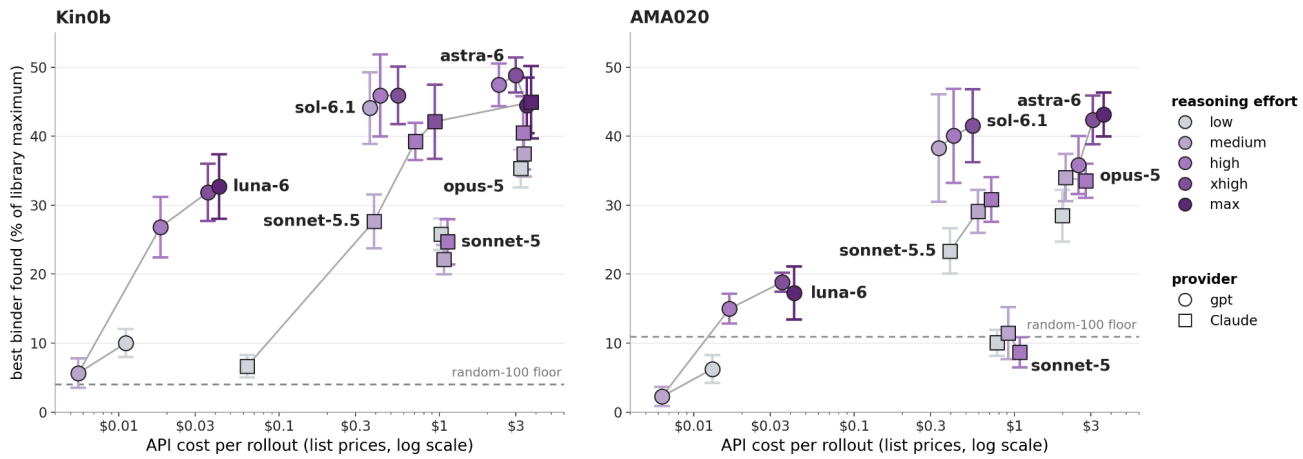


Figure 5. Score against API cost on Kin0b (left) and AMA020 (right). Each point is one model at one reasoning effort: mean best binder per rollout (% of library maximum) against mean API cost per rollout at list prices. Bars are 95% CIs over 10 replicate sweeps; lines join one model's effort settings. Dashed line: best score from 100 random molecules.

Using the identification of a binder (enrichment score > 2.5) within budget as a proxy for success, we see different models and token spends yield different levels of performance (Figure 5, Figure S1). Agent performance varied substantially across model and inference-effort configurations, with many configurations outperforming random search. Within model families, greater inference effort generally improved performance, although spend alone did not determine success.

Agents learn from structure-activity relationship feedback

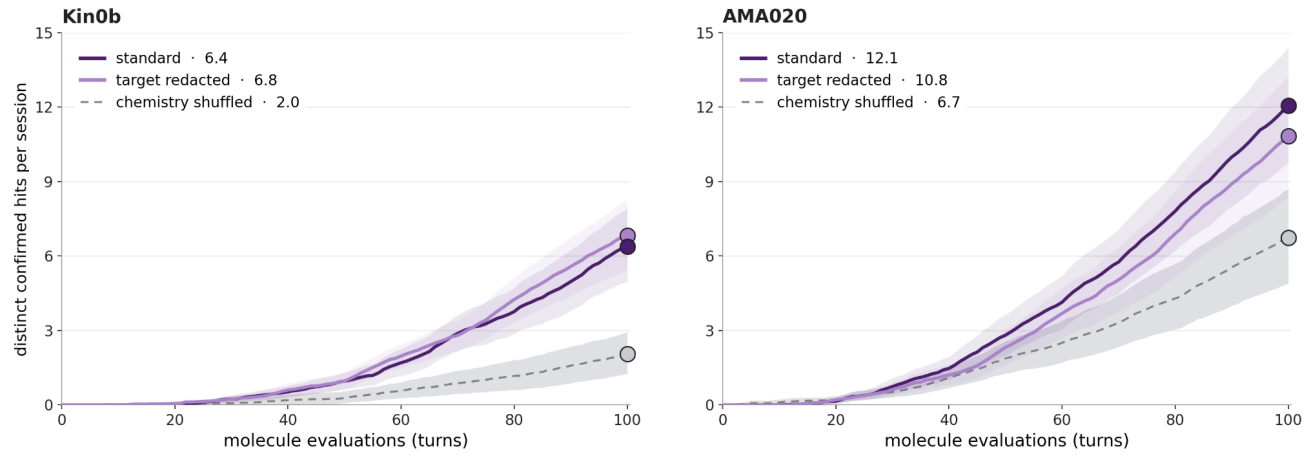


Figure 6. Models use chemistry information; prior target knowledge rarely helps. Distinct confirmed hits per session were measured to follow progress over time. Data are from Opus 5 cold-start sessions on Kin0b (left) and AMA020 (right). Standard: unmodified task. Target redacted: target name hidden. Chemistry shuffled: building-block structures scrambled, IDs kept. Lines are equal-protein means over 10 proteins (10 sessions per protein per condition); bands are 95% bootstrap intervals.

Modern large language models are trained on large corpora of publicly available data, including drug discovery literature. To test the possibility that knowledge obtained from this training might inform SAR tasks in Minos environments, we reran cold-start SAR tasks in 10 different Minos environments across a number of agent-inference effort configurations where we ablated the target name from the task. While there was target-to-target variation, overall we observed little drop in cold-start SAR task performance (Figure 6, Figure S2).

RESULTS / CONTINUED

Because these libraries are combinatorial, improvement on the task need not necessarily imply chemical reasoning. Once an agent encounters promising material, it can in principle treat building-block identities as categorical variables, retain components associated with favorable outcomes, and iteratively replace the others. Such a strategy resembles Mastermind-style local search and could produce progress without interpreting molecular structure. To test this possibility, we scrambled the molecular structures associated with building-block identities while preserving the combinatorial action space and measured outcomes. This retained the possibility of structure-blind local search while disrupting coherent relationships between represented chemistry and physical outcome. After this shuffling, SAR task performance in both AMA020 and Kin0b Minos environments dropped sharply (Figure 6, Figure S2). Indeed, in transcripts from such runs, models struggled. From one transcript:

> **Agent transcript**
>
> **Chemistry priors were anti-correlated.** I front-loaded canonical bromodomain acetyl-lysine mimetics (N-methyl lactams 2339/2058, dimethylisoxazole, methylpyridone). They were the *worst* molecules in the run (2058 → −2.39, the run's minimum). I abandoned structure-based reasoning after ~20 molecules and went empirical; that was the right call but cost me budget.

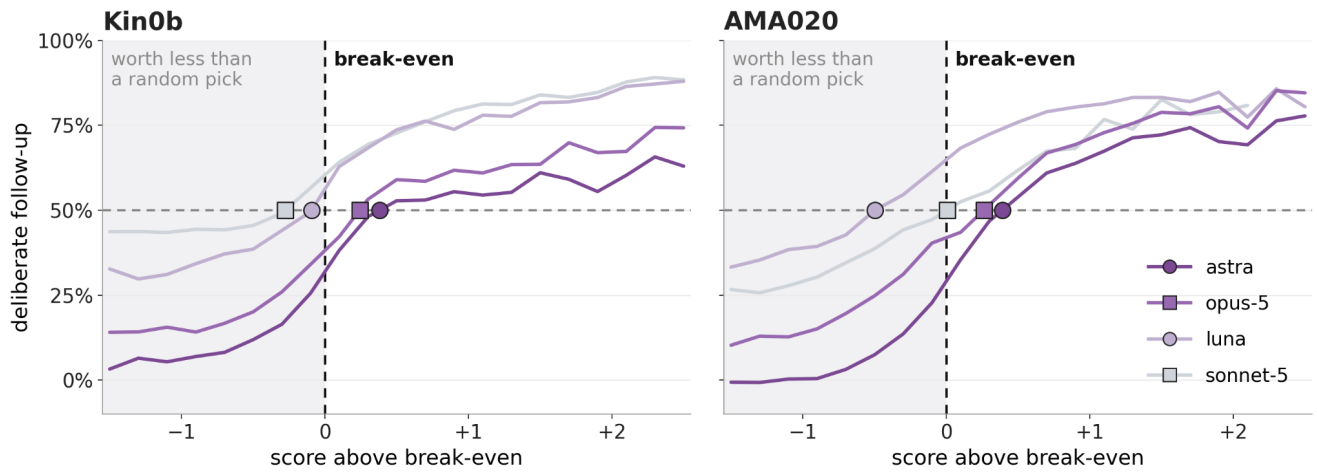


Figure 7. Less performant models are more likely to perseverate in unproductive areas of chemical space. For each tested molecule, we show the probability that the session tests an analogue of it (two of three building blocks shared) within the next three batches, corrected for chance overlap using a different session on the same target. The x-axis is the molecule's score relative to its target's break-even, the score at which analogues hit as often as random picks; the shaded region is below break-even. Markers show where each model's follow-up crosses 50%. Efforts pooled within each model.

Agent decision quality varies across capability

Observing choices made on a limited budget in a fully measured landscape allows for the evaluation of decision-making quality. One way to evaluate such quality is to ask how agent decisions progress towards the best possible outcome in a given Minos environment. A common choice to make in any explore/exploit scenario is the decision to do one or the other; in our SAR chemistry task we defined "exploit" as "choosing molecules with 2 shared building blocks within 3 experimental batches", indicating that an agent is residing in a local chemical neighborhood. By contrast, "explore" is "choosing molecules with 1 or fewer shared building blocks", meaning the agent has moved on to a more distant chemical neighborhood. We suspected models might get stuck in local minima, and predicted that weaker models might do so more frequently.

To test this possibility, we calculated the break-even point of the choice between staying and iterating or moving on. Weaker models iterate in poor neighborhoods and are more likely to perseverate locally overall, whereas stronger models are more likely to abandon unproductive neighborhoods (Figure 7).

RESULTS / CONTINUED

Experimental trajectories reveal agentic strategies

Agents report their findings during runs. Below, we provide a handful of quotations from a single Opus-5 transcript. Emojis are the agent's. The agent identifies promising chemical series, performs truncate tests, identifies binding cliffs, and sketches out a key pharmacophore. These statements represent the agent's interpretation of the measurements; only the experimental outcomes themselves constitute ground truth. We provide a number of complete transcripts at strelka.leash.bio.

Agent transcript

No hits yet, but molecule 1 (bb1=2112, aminopyrrolidinyl-pyridine scaffold) had notably higher target counts (11 vs 1-3). Let me scan that bb1 family.

[...]

Clear SAR emerging: bb1=2410 scaffold + bb2=2653 (2-methoxyquinolin-3-yl) is the driver. Let me find bb2 analogs of that motif.

[...]

Big finding — bb3 is essential (truncates $\approx$ 0) and bb3=1962 (6-(benzimidazol-1-yl)nicotinic acid) jumped to 94 counts. Let me find bb3 analogs of this extended flat motif.

[...]

🎯 First confirmed hit! (2410, 2653, 2336), score 2.71, hit=1, 148 target counts, low bead. bb3=2336 is a 1-methyl-4-oxo-cinnoline-3-carboxylic acid — the amide NH can form an intramolecular H-bond to the 4-oxo, creating a rigid planar pseudo-tricycle.

[...]

Hypothesis confirmed — 4 hits, all sharing a carboxamide ortho to a ring C=O (intramolecular H-bond $\rightarrow$ rigid planar pseudo-tricycle). Best: bb3=1316 at 3.05. The CH_2-spacered uracil (2304) failed, confirming direct attachment is required.

[...]

Dramatic result: with the good bb3, every alternative bb2 collapses to ~0.5 while 2653 gives 3.05 — a very sharp, specific SAR. bb2=2653 is locked.

[...]

🚀 Major jump: bb3=2792 (1-methyl-6-cyclopropyl-pyrazolo[3,4-b]pyridine-4-carboxylic acid) $\rightarrow$ score 3.41, 450 target counts vs 121 library counts. Note its isomer 2158 (cyclopropyl at C3 instead of C6) only gave 1.73 — the C6 substituent is critical.

[...]

Now I understand the full pharmacophore. All three positions are sharply defined (bb1=2410, bb2=2653 are each ~1.5–2.5 log better than any alternative). Let me spend remaining budget hunting bb3 improvements.

The excerpt above was taken from sequential model iterations on a single chemical series. This run produced a binder with a score of 71% of the maximum binder in the Minos world.

Experimental trajectories reveal distinct taxonomies of failure

At every decision point Minos allows for the interrogation of all possible choices in the measured world. This difference between the choice, the agent's stated reason for making that choice, the physical result of making the choice, and the physical results of choices the agent could have made but did not allows for the broad discovery of decision failure modes.

To identify such weaknesses, we examined some 4,602 independent structure-activity relationship trajectories produced by different agent-inference effort configurations. These provided coherent decision failure modes, including stopping before the budget was exhausted or repeatedly testing the same molecule. A fuller accounting of these failure modes can be found in Figure S3. We provide this entire corpus of transcripts at strelka.leash.bio to assist in the identification of a more robust taxonomy of decision failure.

RELATED WORK

Replayable environments have a long history in machine learning, but these have typically been synthetic in order to enable scale, including games such as checkers [12], Atari [13], and Go [14]. In the physical sciences, efforts to provide scientific agents with experience have largely taken one of three forms: 1) reconstructing trajectories through legacy datasets or the literature [3], 2) synthetic environments with computational verifiers [4], or 3) agents directing new experiments in lab-in-the-loop systems [5]. These approaches make different tradeoffs. Legacy datasets are fast and physically grounded, but contain only the experiments their original creators chose to perform. Simulated environments are cheap, fast, and permit broad exploration, but their outcomes are generated by computational approximations to physical reality. Lab-in-the-loop approaches allow agents to choose new physical experiments, but each interaction incurs fiscal and temporal costs, limiting both the number of trajectories that can be generated and access to unchosen counterfactuals.

Minos occupies a different point in this design space. By densely premeasuring large experimental worlds, physical cost and latency are paid during construction and then amortized across subsequent agent interactions. The resulting environments are physically grounded, contain large numbers of counterfactual outcomes, and can be replayed at computational speed without being conditioned on the choices of any particular agent. Scaling the number of independent Minos environments then enables both repeated evaluation of scientific-agent capability and detailed interrogation of agent behavior. The same properties also make these environments natural substrates for reinforcement learning and other forms of post-training, where repeated interaction with physical outcomes can be generated without repeatedly returning to the laboratory.

CONCLUSION

We present Minos, a collection of premeasured experimental worlds based on protein-ligand interactions that amortize the temporal and financial costs of physical experimentation across many agentic rollouts. Scientific agents acting within Minos environments make experimental choices, receive measured physical outcomes, and use those outcomes to inform subsequent choices. Across two independently constructed chemical libraries, agent performance varies substantially across models and inference-effort configurations, while perturbing the correspondence between chemistry and physical outcome impairs performance. Dense measurement also allows us to examine how agents allocate experimental budgets and respond to evidence, revealing differences in scientific search policy that are not visible from endpoint performance alone.

These results suggest that Minos environments are useful both as evaluations of scientific agents and as tools for studying how those agents behave. More broadly, premeasurement provides a way to convert expensive physical experiments into reusable scientific experience. Although we focus here on evaluation and behavioral analysis, replayable physically grounded environments may also provide a scalable substrate for reinforcement learning and other forms of post-training. As many drug-discovery problems are themselves sequential decisions about which experiments to perform and which chemical hypotheses to pursue, these environments provide a direct substrate for training experimental policies. Whether experience acquired in such environments improves scientific decision-making in previously unseen physical worlds is an important question for future work.

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SUPPLEMENTAL FIGURES

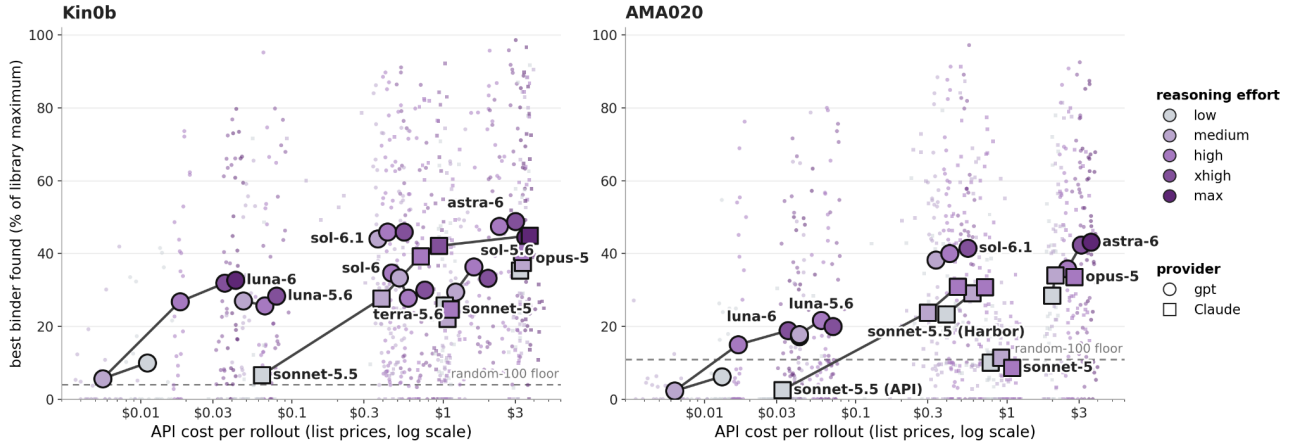


Figure S1. Score against API cost on Kin0b (left) and AMA020 (right). Each point is one model at one reasoning effort: mean best binder per rollout (% of library maximum) against mean API cost per rollout at list prices. Bars are 95% CIs over 10 replicate sweeps; lines join one model's effort settings. Dashed line: best score from 100 random molecules. Dots represent mean individual target performance per model-effort configuration.

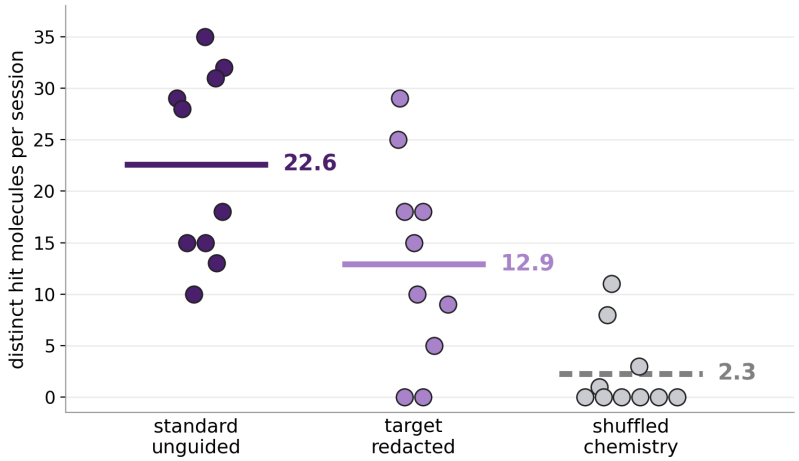


Figure S2. Models use chemistry information; deep prior target knowledge may help. Data are from 10 Opus 5 cold-start sessions per ablation on a BRD9*AMA020 environment. Standard unguided: unmodified task. Target redacted: target name hidden. Shuffled chemistry: building-block structures scrambled, IDs kept.

SUPPLEMENTAL FIGURES / CONTINUED

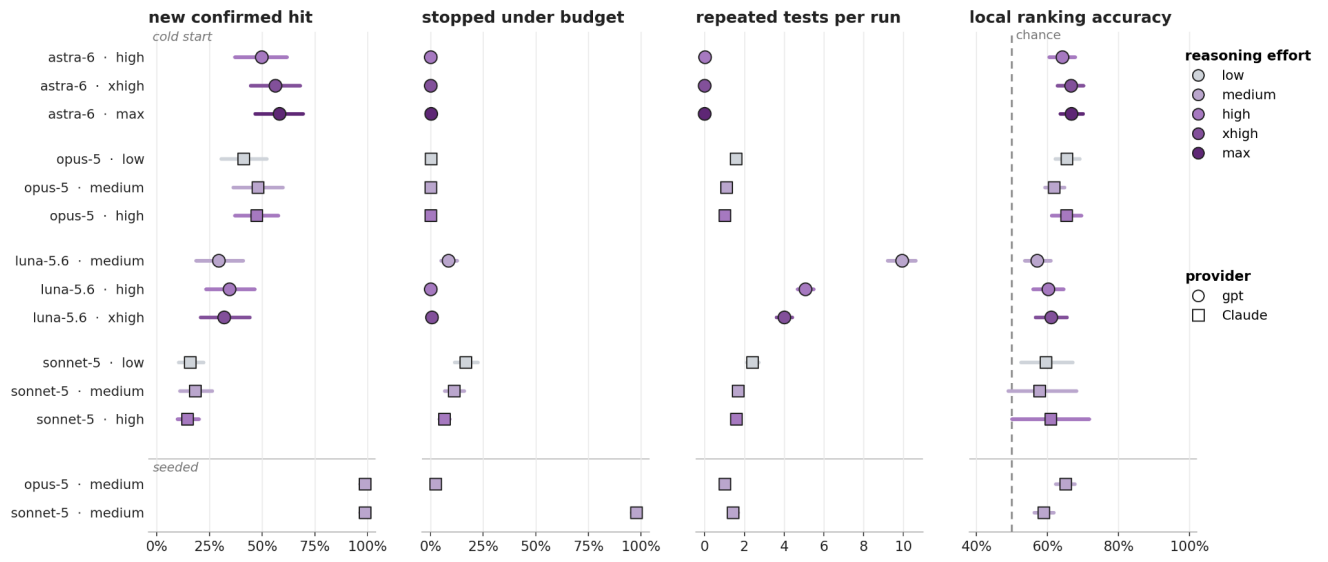


Figure S3. **Same experimental worlds, different observed policies.** Experimental behavior across model families and inference-effort levels in AMA020 environments. Analyses include 4,067 cold-start trajectories across 34 targets and 522 seeded trajectories across 29 targets; each run had a budget of 100 molecule tests. Column 1: Percentage of runs discovering at least one confirmed hit (`hit = 1`) beyond any supplied seed molecules; improvement over seed activity is not required. Column 2: Percentage of runs ending before exhausting the test budget of 100 molecules. Column 3: Mean number of exact repeat molecule tests per run, which needlessly burns budget. Column 4: Post-discovery local ranking accuracy, calculated from predicted versus measured ordering of eligible first-tested, non-seed molecule pairs differing at one building-block position and separated by at least one score unit. Points show means across eligible runs; error bars indicate 95% confidence intervals from 2,000 target-cluster bootstrap resamples. Thirteen cold-start trajectories were excluded for incomplete or inconsistent tool records. Under-budget termination does not establish the reason for stopping, and model-dependent eligibility and molecule selection in column 4 preclude interpreting it as a matched comparison of intrinsic SAR capability.

APPENDIX

Token cost methodology

We estimate API-equivalent inference cost by multiplying recorded token usage by a fixed, model-specific API price schedule. It measures what the observed usage would cost under that schedule; it does not measure subscription spending or quota consumption.

Seed selection vs cold-start methodology

The seed task presents agents with a target and 5 binders to that target sampled from the lower 50% of binders by enrichment score. Agents generate SAR hypotheses on why the 5 binders bind, write SMARTs queries to find building blocks that would help test those hypotheses, and test those hypotheses by interrogating new molecules constructed from those chosen building blocks for binding scores.

We find the seed task easier for competent models; some of this ability is likely attributable to Mastermind-like local search.

Transcript-based analysis of experimental policies

We analyzed 4,602 trajectories from the AMA020 benzimidazole-library benchmark: 4,080 cold-start runs spanning 34 target constructs and 12 model–effort configurations, and 522 seeded runs spanning 29 targets, three seed sets, three replicates, and two models. All runs used unguided task prompts and an allocation of 100 molecule tests. Thirteen cold-start transcripts were excluded because incomplete tool records or inconsistent test counters prevented reliable reconstruction, leaving 4,589 trajectories. Model identity and inference effort were obtained from recorded metadata.

We reconstructed experimental actions by pairing synthesis requests with their corresponding tool responses. Molecules were identified by their ordered building-block triplets (`bb1_id`, `bb2_id`, `bb3_id`). Behavioral metrics were calculated from recorded actions and observations rather than from agents' narrative descriptions. A new confirmed hit was defined as a tested molecule with a returned label of `hit = 1` that was not among the supplied seeds. Each run was scored according to whether it discovered at least one such molecule; exceeding the activity of the supplied seeds was not required. Under-budget termination was defined as a recorded terminal result with fewer molecule tests than allocated. Exact re-query cost was the number of tests whose building-block triplet had already been tested earlier in the same run.

Post-discovery local ranking accuracy was evaluated in synthesis batches submitted after a confirmed hit was available, including supplied seed hits. Eligible comparisons comprised pairs of molecules first tested in that batch, excluding supplied seeds, that differed at exactly one building-block position and had measured scores separated by at least one unit. We compared each pair's ordering in the agent's submitted `predicted_order` with its measured score ordering. Accuracy was calculated within each run and then averaged across runs with at least one eligible pair. Transcripts with ambiguous ordering of synthesis calls and feedback were excluded from this metric. Pair eligibility did not require both molecules to have confirmed-hit labels.

For each model–effort–starting-condition group, point estimates were calculated with equal weight per eligible run. We estimated 95% confidence intervals using 2,000 bootstrap resamples of target constructs, retaining all runs belonging to each sampled target. Denominators were metric-specific, and confidence intervals were not adjusted for multiple comparisons. These analyses describe observed experimental policies: under-budget termination does not establish the reason for stopping, and differences in target eligibility and model-selected molecule pairs limit interpretation of ranking accuracy as a comparison of intrinsic SAR capability.

APPENDIX / CONTINUED

Cold start prompt

Task prompt

You are an AI drug discovery agent exploring a combinatorial molecular library to find the strongest binder for the protein target **PDE10A_S715-D1055**.
You have three MCP tools available: `search_building_blocks`, `synthesize`, and `get_status`. Use these tools directly. Do not attempt to use Bash, curl, or any other tools.

Starting point: no examples

You are starting cold. You are given **no example molecules** and no prior hits — just the library's chemistry, described below. Your only window into the library is `search_building_blocks`: you discover candidate building blocks by SMARTS search, assemble them into molecules, and learn what binds by testing. Nothing is known about this target's hits until you synthesize and measure.

Library structure

The library is fully combinatorial: every `(bb1_id, bb2_id, bb3_id)` triplet is a real, screenable molecule, and the library is the product of the three building-block menus (millions of molecules).
The building-block catalogs are the entire design space — you cannot invent a building block, only discover existing ones via `search_building_blocks`.
The library is large, and most of it does not bind anything. Finding even one confirmed binder is a genuine result.

Scoring

Prioritize molecules with `hit=1` and high **score**. Hits with higher scores are stronger binders, but the measurement is noisy and a difference of less than 1 is not considered significant.

Goal

Find the strongest binder you can within a budget of **100** molecules. Track remaining budget via `budget_remaining` in each `synthesize` response. When your budget runs out, call `get_status` and report your final results.
You also have a separate budget of **200** `search_building_blocks` calls. Each `search_building_blocks` response reports `searches_remaining`; when it hits zero, further searches return an error.

Tools

- `search_building_blocks(position, smarts)` searches BBs at `bb1`, `bb2`, or `bb3` using a SMARTS pattern and returns matching IDs and SMILES. Each response includes `searches_remaining`.
- `synthesize(molecules, predicted_order)` tests up to 5 molecules per batch. `molecules` is a list of `{"bb1_id": ..., "bb2_id": ..., "bb3_id": ...}` objects. `predicted_order` is your predicted ranking from best to worst score as 1-based indices into the batch.
- `get_status()` returns run summary, both budgets, best hits, and recent history.

Background

You are searching a DNA-Encoded Library (DEL). Each molecule is defined by three building blocks: `bb1`, `bb2`, and `bb3`. The library contains millions of molecules screened for binding.

Data interpretation

- `score` is the bead-corrected binding score. Higher is better. Scores can be negative.
- `counts_bead` is the non-specific binding control. It is already accounted for in the score, and high bead counts indicate noise unrelated to the protein.
- `hit` is 1 (hit), 0 (miss), or -1 (dropped, no confident label).

Chemistry context

- `[Dy]` in SMILES is the DNA tag attachment. Ignore it for SAR.
- `bb1` molecules are Boc-protected amines with carboxylic acids or halides.
- `bb2` molecules are boronic acids/esters from Suzuki coupling chemistry.
- `bb3` molecules are carboxylic acids from amide coupling chemistry.

Truncates

- `bb3_id=173` (SMILES: `C`) means no bb3 was added.
- `bb3_id=15012` (SMILES: `[Es]`) means Fmoc deprotection only.

Rules

- Discover BBs only via SMARTS search.
- Test molecules as `(bb1_id, bb2_id, bb3_id)` triplets.

Now begin.